

Many-Inbetweens Browsing: Exploring Pre-Generated Inbetweens for 2D Animation

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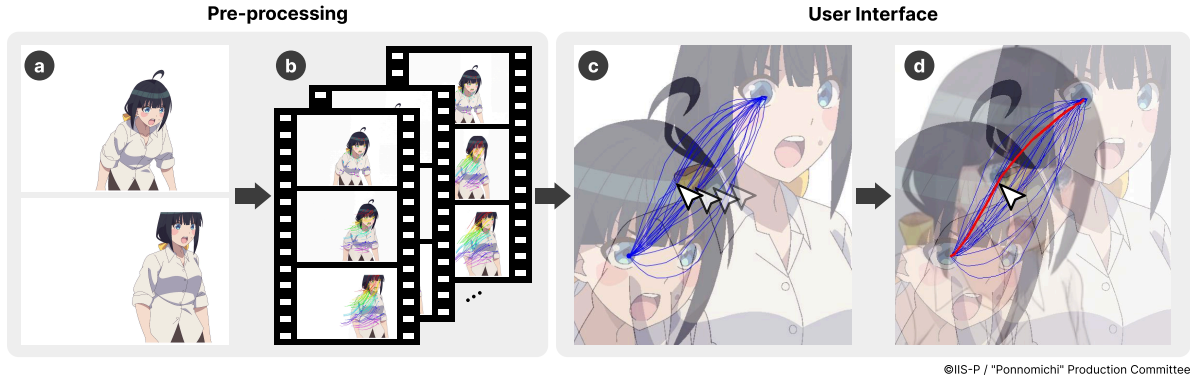


Figure 1: System overview. (a) Given start and end keyframes; (b) a diverse set of inbetweens is generated using a video generation model [Zhang and Agrawala 2025], and their motion trajectories are derived using a point tracking method [Karaev et al. 2024]. This approach is based on professional inbetweening practices, where animators employ motion curves to represent trajectories and timings. (c) In the user interface, the system displays the extracted point trajectories with the blended keyframes; (d) the user can select a trajectory and browse the corresponding inbetween in real time.

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1 Introduction

In 2D animation production, inbetween frames are created to interpolate between given keyframes. This process is time- and labor-intensive, requiring iterative trials and validations. Recent advances in video generation models have provided a new method that simplifies the inbetweening task; these models can synthesize natural

inbetweens using given keyframes [Xing et al. 2024; Zhang and Agrawala 2025]. High-quality inbetweens can be used as final outputs, requiring only minor refinements; low-quality inbetweens still provide useful visual references that help animators to anticipate motion and spatial arrangement. In both cases, video generation models can substantially reduce the time and effort required to create inbetweens.

User experience is highly important when applying video diffusion models to generate inbetweens. A video generation model can readily produce diverse results by varying random seeds. However, the start and end keyframes alone cannot uniquely determine a motion, leading to multiple plausible inbetweens. As a result, users must go through a tedious search process to find the inbetween of interest from a large number of candidate inbetweens.

To address this issue, we propose a system that enables efficient exploration of various pre-generated inbetweens (Figure 1). Our method is based on previous systems that support interactive selection from a diverse set of samples [Pandey et al. 2023; Twigg and James 2007]. Given the start and end keyframes, we

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generate a large set of candidate inbetweens using a video generation model [Zhang and Agrawala 2025]. Based on professional inbetweening workflows, where animators use motion curves to represent trajectories and timings, we construct feature points and their motion trajectories using a point-tracking method [Karaev et al. 2024]. Our interface allows users to select the trajectories and view the corresponding inbetweens in real time, assisting them in exploring various inbetweens. This approach helps users to efficiently identify inbetweens that align with their creative intentions.

2 System Overview

2.1 Inbetween Pre-generation

We employ a video generation model proposed by [Zhang and Agrawala 2025] to generate inbetweens using given start and end keyframes, although any model can be applied. During generation, we specify the text prompt “Anime Style” to obtain a 2D animation appearance. Furthermore, we randomly initialize the model by sampling noise from a normal distribution with varying seeds. We generated 50 variations of inbetween sequences; over two and a half hours; each of them consisted of 37 frames.

2.2 Motion Curves for Inbetween Exploration

We use the scale-invariant feature transform (SIFT) [Lowe 2004] key-points as tracked feature points obtained from the start keyframe; these are selected according to their difference of Gaussians response magnitude because high-contrast areas enable reliable and stable tracking. Then, we apply CoTracker3 [Karaev et al. 2024] to each pre-generated inbetween sequence to obtain the point trajectories; these were treated as motion curves for inbetween exploration. Exploiting these trajectories enables us to efficiently explore and select the motion patterns of interest.

2.3 User Interface

Initially, the system displays the blended keyframes. When users select a feature point, the corresponding trajectories are revealed (Figure 2a). By selecting multiple points, users can independently explore motions at various image parts. When users hover over a trajectory, the system plays back the associated inbetween sequence (Figure 2b). The trajectories are projected onto the first principal component and sorted accordingly for efficient browsing, allowing a transition using the mouse wheel. Additionally, users can press the left mouse button to browse the inbetween associated with the hovered trajectory frame by frame (Figure 2c).

To improve reliability, we filter the generated inbetweens according to their feature point trajectories. Considering that all inbetweens are constrained to match the end keyframe, the trajectory endpoints are expected to coincide. However, as shown in Figure 3a, some endpoints deviate. To overcome this problem, we cluster the endpoints using the density-based spatial clustering of applications with noise (DBSCAN) algorithm [Ester et al. 1996] and retain only those contained in the largest cluster (Figure 3b); this ensures that only consistent inbetweens are presented.

Based on user interaction, our system improves exploration by reordering the trajectories. For each trajectory, the system identifies the point that is closest to the mouse position and applies the

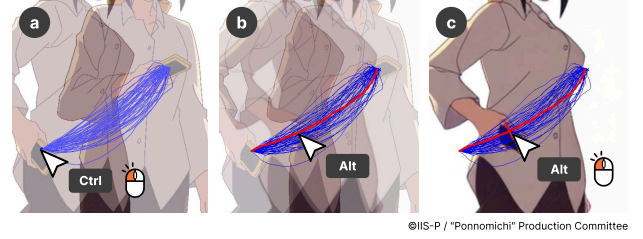


Figure 2: User interface. (a) The user selects a feature point to reveal its trajectories; (b) hovering over a trajectory triggers a playback of the corresponding inbetween; (c) pressing the mouse button converts a trajectory into a slider for frame-by-frame exploration.

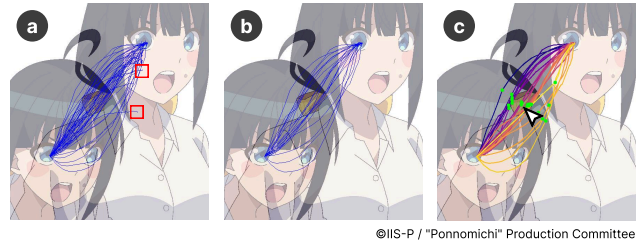


Figure 3: Trajectory processing. (a) Extracted trajectories before filtering, showing the endpoint outliers (red); (b) consistent trajectories obtained by clustering the endpoints using the DBSCAN algorithm; (c) sorted trajectories and trajectory points used for sorting (green) (the color of a trajectory indicates its order).

principal component analysis algorithm to determine the principal direction. By sorting the trajectories along this direction, the previously unstructured set is transformed into an ordered representation (Figure 3c).

3 Expert Interviews

As a preliminary evaluation, we asked an animator with over seven years of experience to use our system and provide feedback. The animator responded positively regarding the ability to browse various inbetween examples in real time. Additionally, compared with conventional video generation models, where prompt-based control often fails to capture user intention, our system allows users to select inbetweens that align with the motion trajectories of interest more easily. However, the animator pointed out that the current system does not provide fine control over individual image parts, such as the face or hands, making it difficult to fully implement creative intentions.

4 Future Work

Based on the feedback obtained from the animator, our future objective is to allow users to specify parts that trajectories must pass through, enabling a detailed adjustment of the inbetween motions. We anticipate that this will allow creative intentions to be more accurately reflected during inbetween exploration.

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